History and Chunking Under Partial Observation

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Abstract

We compare history and action commitment in the linear hard example. Recovery training with history two favors single-action feedback. Narrow training can still benefit from chunking despite observability. Without history, every tested learner remains unstable. These results separate information, training coverage, and feedback stability.

1. Design and evaluation

The hard example of Simchowitz et al. (2025) pairs plants A_1, A_2 with experts K_1, K_2 that generate identical demonstrations on $x_1 = 0$ but require different recovery actions. We fix A_1 at $\mu = 1/8$ and expert K_1. Both plant and expert loop are stable, yet K_2 matches the narrow demonstrations and destabilizes A_1 after a departure: training fit need not identify safe recovery.

All runs observe only $y = x_2$. History r contains r observations and $r - 1$ applied actions; no history means $r = 1$. The four conditions cross history $r \in \{2, 4, 8, 16\}$ versus $r = 1$ with chunked versus single-action prediction. Chunking predicts $H = 16$ actions and executes $\ell \in \{2, 4, 8, 16\}$ before replanning; no chunking uses $H = \ell = 1$.

Both narrow training ($x_1 = 0$) and recovery training use ridge regression ($\lambda = 10^{-6}$). Recovery collection applies known probes $u = K_1 x + 0.05\xi$, $\xi \sim \mathcal{N}(0, I)$, labeling visited histories with nominal expert chunks $K_1 F^j x$, $F = A_1 + K_1$. Inputs retain applied actions. This adapts noisy collection and expert correction (Laskey et al., 2017; Ross et al., 2011), motivated by Zhang et al. (2025), without implementing those algorithms in full.

Seeds 818–820 each supply 32 trajectories of 80 steps; target times match across r, H. The 60 fits yield 150 four-condition records plus 30 $H = 16, \ell = 1$ controls, all numerically matching their single-action counterparts. Thus execution comparisons isolate commitment rather than extra prediction heads.

Test seed 20261107 generates 20 paired states with x_2 uniform on $[-1, 1]$ and $x_1 = 0$ or a signed departure of magnitude 0.0005–0.0015. Following a common 15-step expert warm-up, we inject the departure and run the learner independently for 200 noiseless steps. A separate expert rollout supplies the reference. Mean MSE averages squared full-state tracking distance over post-action states, perturbed cases, and seeds, retaining unstable losses. All 180 seed-level means reconcile with raw rollout losses. Spectral rates below one indicate decay; 1000-step canonical rollouts are separate diagnostics.

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2. Results and analysis

Training	History r	Execution ℓ	Mean test MSE	Worst rate
Narrow	1	16	5.19×10^{-3}	1.02547
Narrow	8	1	7.58×10^{-4}	1.01892
Narrow	8	8	2.32×10^{-6}	0.99544
Narrow	16	1	6.65×10^{-7}	0.97110
Recovery	1	16	2.53×10^{-1}	1.02618
Recovery	2	1	7.33×10^{-9}	0.75000
Recovery	2	16	2.71×10^{-7}	0.75000

Rows with $\ell = 1$ use $H = 1$; others use $H = 16$. All seeds agree on stability in all 50 cells.

Without history, every fitted learner remains unstable. Narrow loss falls from 9.06×10^{12} at $\ell = 1$ to 5.19×10^{-3} at $\ell = 16$, yet the latter has 1000-step canonical MSE 4.08×10^{14}: commitment slows divergence without stabilizing. This is not an impossibility result; zero control stabilizes the plant.

With recovery data, all $r \geq 2$ settings are stable. History two with one-step feedback nearly matches the model-informed observer (7.334×10^{-9}). Executing two or sixteen actions increases MSE 2.30- or 36.9-fold: commitment delays correction. Longer history also need not help: one-step MSE reaches 3.04×10^{-8} at $r = 16$, with rates still near 0.75.

With narrow data, observable history two still fails at every execution. History eight is stable only at $\ell = 8, 16$; execution eight reduces MSE 327-fold. History sixteen is stable throughout, with only 11% improvement from execution sixteen over one. Extra history changes ridge-selected feedback without adding corrective examples: observability alone does not eliminate chunking benefits.

3. Connection to Kyle

Kyle's reconstruction result explains the history-two threshold. The hidden departure becomes observable after one transition. Near-observer recovery performance supports reconstruction; narrow-data failure separates representation from identification.

His chunk-stability condition requires $\ell \geq r - 1$, excluding stable $(16, 1)$; its Lipschitz assumptions remain unchecked. The loss corollary needs reference-deployment error, not training fit, and bounds output rather than full-state error. These experiments illustrate mechanisms, not verification of the guarantee.

Context plus corrective coverage removes the observed commitment benefit here; observability alone does not. Three seeds, a stable plant, small departures, expert initialization, and amplitude shift limit generalization. Finite-rollout loss must be read alongside spectral stability.

References

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